PROPORTION, KNOWLEDGE AND SELECTED ASSOCIATED FACTORS OF ANAEMIA AMONG NON-PREGNANT FEMALES OF REPRODUCTIVE AGE IN A TEA ESTATE COMMUNITY IN HANTANA, KANDY DISTRICT, SRI LANKA

Management and Science University

Abstract

Ministry of Finance and Mass Media in 2018 shows that there is a high prevalence of anaemia in Kandy and the government of Sri Lanka has issued a cabinet decision in June 2018 stating, “Promoting nutrition of meals by adding Iron and Folic acid to minimize anaemia”. This research is targeting a selected estate population in Kandy district to assess the proportion, knowledge and selected associated factors of anaemia among non-pregnant females of reproductive age. A descriptive cross-sectional study will be conducted in a tea estate community in Hanthana, Kandy District. Main purpose of the study is to find the current proportion of anaemia in the study population and find its corresponding associated risk factors so necessary interventions can be made to better the lives of the tea estate community since estate communities are being identified as risk populations to many deficient health conditions. The results can be generalised to other tea estate communities across the country so further studies can be conducted in the future. A total of 330 tea estate dwellers will be participating in this research where the sample size was calculated in relevance to the current prevalence of anaemia in Kandy district which is 29%. After obtaining the written informed consent from the volunteer participants, their blood sample will be drawn by a trained phlebotomist to check their full blood count and level of haemoglobin to assess the anaemic level. Then their height and weight will be measured using height scale and a weighing scale respectively, an equation will then be used to calculate their BMI. Finally an interviewer administered questionnaire together with a 24-hour dietary recall will be administered to evaluate the associated factors with anaemia in the population. Analysis of data will be done on IBM SPSS Statistics software package version 25 where descriptive statistics, T-test, chi-square and uni-variant analysis will be used. Proportion of anaemia will be given at a confidence interval of 95%. Ethical issues that may imply in this research will be the written consent for collection of data and blood samples from the study participants before conducting the research, maintenance of privacy and confidentiality of the participant’s information. Ethical clearance for the protocol will be obtained from NIHS Kalutara and the administrative clearance prior to the research will be obtained from the Estate Management. Samples will also have to be collected, transported, stored, analysed and discarded according to proper guidelines as given by an international recognized body.

Research proposal

PROPORTION, KNOWLEDGE AND SELECTED ASSOCIATED FACTORSOF ANAEMIA AMONG NON-PREGNANT FEMALES OF REPRODUCTIVE AGEIN ATEA ESTATE COMMUNITY IN HANTANA, KANDY DISTRICT, SRI LANKA

1. Introduction

1.1 Background information

Anaemia is a health issue across the globe that is a result of decreased total mass of the red blood cells (RBCs) below the reference range given by World Health Organization (WHO), which results in the inadequacy of oxygen to meet the body’s needs (Cappellini & Motta, 2015). General cut-off level for the diagnosis of anaemia is 12 g/dL of haemoglobin (Hb) for a non-pregnant female who is 15 years and above. In contrast, males of similar age have a higher cut-off value of 13 g/dL (WHO, 2011). The cut- off of 12 g/dL applies to only non-African, non-pregnant, female, older than 15 years residing in an altitude less than 1000m (Sullivan *et al*., 2008). Certain factors such as gender, altitude, age, ethnicity, pregnancy, smoking behaviour and geography may alter the level of the normal cut-off mentioned above (WHO, 2011; WHO, 2017). Anaemia can be further classified as severe, moderate and mild depending on the level of Hb in the blood. Lowered mass of erythrocytes can be caused by three main biological mechanisms namely: low RBC count due to ineffective erythropoiesis, extreme blood loss, increased destruction of blood cells due to haemolysis or a combination of these factors (National Heart, Lung and Blood Institute, 2016; WHO, 2017).

Anaemia is a multi-factorial condition with the most common form of causation being nutritional anaemia followed by diseases (infection and inflammation) and genetic blood disorders (Petry *et al*., 1989). Nutritional anaemia occurs when the body’s requirement of nutrient is compromised. Iron Deficiency Anaemia (IDA) is a major form of nutritional anaemia which victimise 50% and 42% respectively of all worldwide non-pregnant/pregnant women and children under 5 years who have anaemia (WHO, 2017). Deficiencies of folate, copper, pyridoxine, riboflavin and vitamins such as A, B, C, D and E can cause nutritional anaemia (Gupta *et al*., 2016; WHO, 2017). Moreover, communicable disease like intestinal parasitic infections, tuberculosis, malaria and schistosomiasis can cause anaemia through numerous mechanisms (Petry *et al*., 1989; Gilgen & Rosetta, 2001). Patients with chronic Non-communicable Diseases (NCD) like cancer, heart failure and rheumatoid arthritis suffer critically from anaemia (Anaemia of chronic disorder) and this is said to be second to IDA in terms of its prevalence. Genetic blood disorders due to impaired Hb, membrane and enzyme like sickle cell anaemia are prevailing in the sub-Saharan Africa proceeded by thalassemia which is of high prevalence in countries south-east Asian countries like Thailand (Low *et al*., 2016).

Anaemia can be also classified considering the size, shape and colour of the RBCs. According to this cytometric classification there are three main groups namely; hypochromic microcytic, normochromic macrocytic and normochromic normocytic anaemia. The size of cells smaller than 80 fl are considered as microcytes (Janz, Johnson & Rubenstein, 2013).

It is estimated that around 24.8% of the world's population suffers from anaemia (WHO/CDC, 2005). Among this, a total of 468 million are non-pregnant females while 56 million are pregnant females. Haemoglobin bench mark for anaemia differs from country to country, for example in United States the value is adjusted for smoking while in Laos it was adjusted for altitude (Wirth *et al.,* 2017). Burden of anaemia in a country is classified into mild, moderate and severe and their respective prevalence's of anaemia being 5.0% - 19.9%, 20.0% - 39.9% and above 40% (Wirth *et al.,* 2017). If a country maintains prevalence less than 5% of anaemia, it is considered normal. Statistics show that 50% of anaemia is caused by IDAs making it the most common nutritional deficiency in the world (Low *et al*., 2016). Iron deficiency anaemia is widely prevalent in countries like Liberia (21.2%) and Mozambique (17.3%). In the Asian context the prevalence of anaemia in children, non-pregnant women and pregnant women is said to be 42.0%, 31.6% and 39.3% respectively (WHO, 2011). Nutritional anaemia is on the red alert among the females of India; the causative factors are low intake of iron together with infections (Trakroo, Mahanta & Mahanta, 2013). In addition to that there is a 30% prevalence of anaemia in non-pregnant females (Pickers *et al*., 2014). Proportion of the population having Hb levels less than 110g/L in India is 74.3 with a 95% CI of 73.4, 75.1 and in Sri Lanka it is 29.9 with a 95% CI of 27.0-33.0 (WHO, 2005).

Risk factors of anaemia can be categorized as proximal and distal factors. Proximal factors are those that are directly responsible for causing the health condition and in this case iron deficiency; low levels of vitamin A, imbalanced body mass index, inflammation and infections such as malaria are a few examples. Education, Water, sanitation and hygiene (WASH) and demography are some of the identified distal risk factors (Wirth *et al*., 2017). Infants, young children, women of reproductive age, pregnant mothers and older adults are more vulnerable to anaemia (WHO, 2017). Anaemia has a 3.9% prevalence among children from the age of 1-5 (Gupta *et al*., 2016; Food and Agriculture Organization, 2017). Anaemia in women of reproductive age is increasing in an alarming rate (33% prevalence) which explains the Sustainable Development Goal 2.2 that emphasises the nutritional needs of lactating, pregnant and girls who have reached adolescence (FAO, 2017). Due to the loss of blood during menstruation, women are at higher risk when they are in the phase before menopause but after the onset of menarche (Low *et al*., 2016). Adolescent girls have a higher requirement of iron due to growth thus needing more iron than their male counterparts. Research shows that with advancing age the prevalence of anaemia increases specially after the age of 50 (Patel, 2008). Symptoms of anaemia during its onset phase are not quite definite, varying from shortness of breath, fatigue, weakness and pallor (Low *et al*., 2016). It significantly lowers productivity, cognitive skills and immuno-competence. The effects are severe in old aged people as they can fall prey to dementia, disability and also could lead to death (WHO, 2017).

According to WHO the haemoglobin cut-off for anaemia is Sri Lanka 12.0 g/L. However, since the research is conducted in Hantana which is approximately 1300 m above normal sea level, an altitude correction is made as mentioned in the WHO guidelines (WHO, 2011). Therefore the cut-off in this study is 12.5 g/dl , except for the individuals that smoke (Sullivan *et al*., 2008). The prevalence of anaemia in Sri Lanka among non-pregnant females between the ages 15-49 years is around 32.6% according to WHO (2016). Demographic & Health Survey suggests 38.1% of non-pregnant women are suffering with anaemia and trends show the prevalence increases with age. A study conducted on female working population who have a low socioeconomic standard of living found that anaemia is of high prevalence in this population (Kalasuramath & Vinod Kumar, 2015). Non-pregnant women in the estate population reported to be having a higher prevalence of anaemia in compared to the urban and rural crowd (Demographic & Health Services, 2006). About 20-29% of the non-pregnant women aged 15-49 years in Kandy are with anaemia (DHS, 2006). It is also suspected that they consume high amounts of tea which is an inhibitor of iron (Low *et al*., 2016).

1.2 Justification

Tea is a major plantation crop in Sri Lanka accounting for around 10% of cultivated acreage (Samarasinghe, 1993). This industry holds up to 6% of the total work force in the country. Female tea-pluckers face a double burden situation, where they have to work in the estate as tea pluckers and manage the household by themselves. Research shows that 77.4% of the women are illiterate therefore giving rise to the need for an interview administered questionnaire. Their level of nutrition is also tampered due to male bias in intra-household food allocation (Samarasinghe, 1993). Data shows that there is a high prevalence of anaemia in Kandy and the government of Sri Lanka has issued cabinet decision in June 2018 stating "Promoting nutrition of meals by adding Iron and Folic acid to minimize anaemia" (Ministry of Finance and Mass Media, 2018).

1.3 Objectives

1.3.1 General objectives

- To determine the proportion, knowledge and selected associated factors on anaemia among non-pregnant females of reproductive age in a tea estate community in Hantana, Kandy district, Sri Lanka.

1.3.2 Specific objectives

- To determine the proportion of anaemia among non-pregnant females of reproductive age in a tea estate community in Hantana, Kandy district, Sri Lanka.
- To determine the knowledge on anaemia among non-pregnant females of reproductive age in a tea estate community in Hantana, Kandy district, Sri Lanka.
- To assess the selected associated factors of anaemia among non-pregnant females of reproductive age in a tea estate community in Hantana, Kandy district, Sri Lanka.

2 Materials and methods

2.1 Study design

Descriptive cross-sectional study

2.3 Study setting

Hantana estate tea factory is the largest estate in Hantana area situated near Uduwela road. It consists of a total population of 3000 tea estate workers. It is situated at an altitude of 1300 m from sea level with a temperature of 30 ºC and humidity of 93%.

2.4 Study period

The research will be conducted from August 2018 to April 2019 for an 8 months period.

2.5 Study population with Inclusion and Exclusion criteria

Study population is the non-pregnant females of reproductive age in a tea estate community in Hantana, Kandy district. This study setting has been chosen due to logistical reasons coupled with the questionable prevalence of anaemia in the region.

Inclusion – Non-pregnant females of reproductive age residing in the estate

Exclusion –Females who are not voluntary to donate their blood sample

2.6 Calculation of sample size

The Lwanga and Lemeshow equation is used to determine the sample size required to conduct this research (Lwanga & Lemeshow, 1991). Confidence interval of this study is set to 95% therefore the standard deviation is 1.96. In Kandy, the prevalence of anaemia is said to be 29% among non-pregnant females according to DHS (2006).

$n = Z^2P(1-P) / d^2$

Where; n = sample size

Z = critical value of specified confidence interval

d = desired precision

P = estimate of population

$n = Z^2P(1-P) / d^2$

$n = 1.96^2 \times 0.29 \times 0.71 / 0.05 \times 0.05$

$n = 316$

2.7 Sampling technique

Simple random sampling technique will be used when selecting the sample population. The sampling frame in this case is the list of female tea estate dwellers which will be extracted from the estate manager where 316 female workers will be randomly, selected using a random number generator.

2.8 Study Instruments

2.8.1 Questionnaire

Pretested interviewer administered questionnaire (Annexure V) will be used together with a 24-hour dietary recall. Questionnaire will be developed in English language and it will be translated to Sinhala and Tamil languages. A trained data collector who speaks the native language will administer this questionnaire to all. Further validation of questionnaire will be revised by a haematologist. The questionnaire is divided into sections as mentioned below;

- Part I – Socio-demography, clinical representation and nutritional assessment
- Part II –Reproductive health
- Part III – Knowledge on anaemia
- Part VI- 24 hour dietary recall

2.8.2 Measure of anthropometry (BMI- Body Mass Index)

A mathematical formula will be used to measure BMI. Weight will be measured using a scale in ‘Kg’ while height will be measured using a tape measure in ‘cm’ and later will be converted to ‘m’ (Eftekhari & Shidfar, 2009). All these measurements will be taken by the principal investigator. The equation given below will be used to calculate BMI.

$\text{BMI} = \text{weight (kg)} / \text{height}^2 \text{ (m}^2\text{)}$

Kg- kilograms; m- meters

Normal range of BMI is from 18.5- 24.9. Overweight 25-29.9 and obesity is more than 30 and underweight <18.5 (Cook *et al*., 2017).

2.8.3 Laboratory methods (Blood haemoglobin measurement and blood picture)

Mindray five-part automated blood analyser will be used to measure RBC parameters such as Hb concentration, haematocrit, WBC (White blood cells) differential count, mean cell haemoglobin (MCH), mean corpuscular volume (MCV), mean corpuscular haemoglobin concentration (MCHC) and RBC count. This blood analyser uses the principle of tri-angle laser scatter, chemical dye and flowcytometry technology to deduce the parameters. A minimum sample volume of 15 μL is required. Blood will also be smeared on a clean slide and stained in Leishman stain according to a set protocol to be visualised under the microscope. The blood picture diagnosis will be done by a haematologist for the purpose of characterising anaemia.

2.9 Study Implementation

Questionnaire will be pretested on a different estate population in the same district. BMI measurement and blood samples will also be collected from people who will be willing to voluntarily participate.

2.10 Data analysis

IBM SPSS Statistics software package version 25 will be used to process and analyse the data. Knowledge will be given scores and based on that it will be assessed as good, moderate and poor knowledge. Descriptive statistics will be used to describe the data set. Chi-square test will be used to determine the relationship between the variables. T-test is the statistical method used to compare the means of two variables such as the age and monthly income and its association in anaemia. Depending on the uni-variant analysis the multi-variant analysis will be carried out. The proportion will be given with a confidence interval (CI). The cut-off for anaemia will be corrected according to altitude and smoking status as mention in WHO (2011).

2.11 Administrative requirement

Permission will be taken from the estate management (Janatha Estates Development Board) and estate doctor prior to the research. Prior approval will also be taken from the Regional Director of Health services and also the MOH in Hantana area before data collection. The mother and child clinic of the estate are the premises to be used for data collection and prior approval will be taken from the mid-wives/nurses working there.

2.12 Ethical issues and clearance

Ethical clearance will be obtained from National Institute of Health Science (NIHS), Kalutara. Information sheet (Annexure III)will be given to all participants and they will be given the choice to participate as well as the choice to withdraw during any given period during the research. Any queries of the participants will be addressed by the researcher. Informed consent forms (Annexure IV) will be presented to all subjects before extracting data from questionnaire, BMI and drawing blood samples. The researcher will perform the research without disrupting the day to day activities in the estate. The social and private life of the workers will not be affected if they refuse to participate in the research. A trained licensed phlebotomist will be hired by the researcher to draw blood from participants. Privacy and Confidentiality of participants will be maintained before, during and after the completion of the research. Principal investigator will have all data bases and questionnaires. All standard precautions and protocols will be followed when collecting, storing and transporting blood samples from Kandy to Colombo and the test will be conducted within 24 hours. Blood samples will be discarded in an appropriate manner within 2 days of collection. After the completion of the research, a health education program will be conducted on the study population on anaemia. All subjects who are suspected in having anaemia will be advised to seek medical attention immediately.

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Annexure I

Time frame

Gantt chart

Activity	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Ap
Literature survey									
Proposal writing									
Ethical clearance									
Sample collection									
Data collection									
Data entry									
Data analysis/Sample experimentation									
Report writing									

Annexure II

Budget

Items estimated	Cost (Rs.)
Travel	5000
Printing	5000
Full blood count	140,000
Blood collection tubes and syringes	10,000
Clearance charges	3000
Miscellaneous	10000
Total	168,000

Annexure III

Information sheet

PROPORTION, KNOWLEDGE AND SELECTED ASSOCIATED FACTORS OF ANAEMIA AMONG NON-PREGNANT FEMALES OF REPRODUCTIVE AGE IN A TEA ESTATE COMMUNITY IN HANTANA, KANDY DISTRICT, SRI LANKA.

Name of Proposal and version

Prevalence and knowledge on anaemia and its selected associated factors among non-pregnant females within the age of 18-49 in a tea estate community in Hantana, Kandy district, Sri Lanka

Introduction

I am Asma Sheriff an undergraduate student, following a degree program of Biomedical Science at Management and Science Institute (MSI). I am doing a research on Anaemia, which is quite widespread in this country. I am going to give you the necessary information and invite you to be part of this research. You are not obliged to give your decision today regarding your participation in the research. Before you decide, you can talk to anyone you feel comfortable with about the research. In case of any queries people feel free to ask me and I will take my time and clarify your doubts. If you have any further doubts you may ask the supervisor.

Purpose of the research

Anaemia is one of the most common disease conditions in this country. A lot of people are unaware that they have this condition; they fail to detect and treat it during its early onset. The main symptoms of anaemia are lethargy and decreased productivity. Each population has its own specific risk factors that will contribute to anaemia. The reason we are doing this research is to find out how many people are affected by anaemia, and to find out why they develop anaemia.

Type of Research

This research will involve a single injection into your arm followed by the measurement of BMI (body mass index). A questionnaire will also have to be filled.

Participant selection

We are inviting all female tea estate workers to participate in this research on anaemia. Everyone in the estate had equal chance of being selected. Your participation would help me in obtaining results that could benefit the estate community.

Voluntary Participation

Your participation in this research is entirely voluntary. You have the choice if you want to participate or not. Regardless of your participation, your day to day work will remain the same and no change will be incurred upon you. If you choose to participate, you will be provided with the results obtained from your tests and measurements which will be of use when treatment for anaemia is required. You can change your mind later and stop participating even if you agreed earlier.

Procedure and Protocol

A. Unfamiliar Procedure

You will first have to complete an interview based questionnaire. After which measurements of your height and weight will be taken. We will take blood from your arm using a syringe and needle. We will take about 2ml of blood to measure your haemoglobin level. The left-over blood samples will be destroyed within 3 days of collection. Samples taken for the research will not be used for any other purpose.

B. Description process

During the course of the research the following will be conducted,

- A few questions will be asked about your general health, hygiene and diet.
- Your height will be measured using a tape while your weight will be measured using a scale.
- A small amount of blood, equal to the volume of a teaspoon will be taken from your arm with a syringe. This blood will be tested for the level of haemoglobin which will indicate if you are suffering from anaemia or not.

Duration

The research will take place over a period of 8 months. During that time we would like to meet you twice within the course of the 8 months for the collection of sample and information. At the end of the 8th month the research will be completed.

Side effects

There might be temporary swelling around the area from which blood was drawn. It is possible that it may cause some problems that we are unaware of. However, we will follow you closely to make sure there are no major side effects.

Risks

By participating in this research it is possible that you will be at risk than you would otherwise be. There is for example a risk that your condition of anaemia may not be detected. If anything unexpected happens during the collection of blood we will provide you with the necessary first aid and medical attention required.

Benefits

If you participate in this research, you will receive a free check on your haemoglobin level. Your participation is likely to help us find the answer to the associated factors that lead to anaemia. It is a benefit to the society because anaemia reduces productivity.

Reimbursements

You will not be given any money or incentives to take part in the research.

Confidentiality

With this research, we are hoping to make better the estate community in terms of a single aspect of health being anaemia. It is possible that if others in the community are aware that you are participating, they may ask you questions. We will not be sharing the identity of the participants of this research. Complete confidentiality will be maintained with all the information that is collected throughout the research. Only the investigator will have all the information about you, any information about you will be replaced by a number or name to make it anonymous. A lock and key will be used to lock your information in its encrypted form. Your information will not be disclosed to anyone but the principle investigator.

Sharing the Results

The knowledge that we get from doing this research will be shared with you through a community meeting before publishing the research. Confidential information will not be shared. We will then publish the results for the interest of other people who might be interested in our research.

Right to Refuse or Withdraw

You do not have to take part in this research if you do not wish to do so and refusing to participate will not affect your life or work in anyway. You will still have all the benefits that you would otherwise have at the estate. You may stop participating in the research at any time that you wish without losing any of your rights as a participant here.

Alternatives to participating

If you do not wish to take part in the research you can proceed with your day to day work at the estate.

Who to Contact

If you have any questions you may ask them now or later, even after the study has started. If you wish to ask any questions later, you may contact any of the following:

Asma Sheriff – 0765342831 Kalpani Bandaranayaka- 0759360565

This proposal has been reviewed and approved by NIHS Kalutara, which is a committee whose task it is to make sure that research participants are protected from harm. If you wish to find about more about them, contact +94 34 2222264 or +94 34 2222001. You can ask me any more questions about any part of the research study, if you wish to. Do you have any questions?

Annexure IV

Consent form

PROPORTION, KNOWLEDGE AND SELECTED ASSOCIATED FACTORS OF ANAEMIA AMONG NON-PREGNANT FEMALES OF REPRODUCTIVE AGE IN A TEA ESTATE COMMUNITY IN HANTANA, KANDY DISTRICT, SRI LANKA

To be completed by the participant
The participant should complete the whole of this sheet himself/ herself. (Put a circle at your choice)

1. Have you read the information sheet? (Please keep a copy for yourself)
   YES/ NO
2. Have you had an opportunity to discuss this study and ask any questions?
   YES/ NO
3. Have you had satisfactory answers to all your questions?
   YES/ NO
4. Have you received enough information about the study?
   YES/ NO
5. Who explained the study to you? ………………………...
6. Do you understand that you are free to withdraw from the study at any time, without giving a reason?
   YES/ NO
7. Have you had sufficient time to come into a decision?
   YES/ NO
8. Do you agree to take part in this study?
   YES/ NO
9. Do you understand that blood will be drawn from you
   YES/NO
10. Are you willing to allow blood to be drawn
    YES/NO

Participant’s signature……………………………
Date …………….....
Name (BLOCK CAPITALS)……………………………………………………………..

To be completed by the investigator: I have explained the study to the above volunteer and he/ she has indicated her willingness to take part.
Signature of the investigator ……………………………….
Date ………………………..
Name (BLOCK CAPITALS)………………………………………………………………

I have been present while the procedure has been explained to the participant and I have witnessed his/her consent to take part in the study.
Signature of witness …………………………………………
(The witness should be person NOT connected with the study)
Date
Name (BLOCK CAPITALS) ……………………………………………………….

ANNEXURE V
Questionnaire

PROPORTION, KNOWLEDGE AND SELECTED ASSOCIATED FACTORS OF ANAEMIA AMONG NON-PREGNANT FEMALES OF REPRODUCTIVE AGE IN A TEA ESTATE COMMUNITY IN HANTANA, KANDY DISTRICT, SRI LANKA

Date of the interview:

Indicate the correct answer to the following questions by marking "✓" or by filling it in the space provided.

Part I- Socio-demographic, clinical characteristics and nutritional assessment

1. What is your age?	2. How many children do you have? Son ………. Daughter ……….
3. What is your marital status? a. Married ☐ b. Unmarried ☐ c. Divorced ☐ d. Widowed ☐	4. Were your parents related before marriage? a. Yes ☐ b. No ☐ c. I don't know ☐
5. What is your employment status? a. House wife ☐ b. Estate worker ☐ c. Factory worker ☐ d. Other ☐	6. What is your religion? a. Buddhist ☐ b. Christian/ Catholic ☐ c. Islam ☐ d. Hindu ☐ e. Others ☐
7. What is your monthly family income? a. > Rs. 10,000 ☐ b. Rs. 10,000 – Rs. 20,000 ☐ c. Rs. 20,000 – Rs. 30,000 ☐ d. < Rs. 30,000 ☐	8. How many people are residing in your household? ………………………………………
9. What is the highest level of education Have you achieved? a. No formal education ☐ b. Primary education ☐ c. Ordinary level (O/L) ☐ d. Advanced level (A/L) ☐ e. University level ☐	10. Do you smoke? a. Yes ☐ b. No ☐ 11. Do you consume alcohol? a. Yes ☐ b. No ☐
12. Do you have the same type of meals every day? a. Yes ☐ b. No ☐	13. Which category do you belong to? a. Vegan ☐ b. Vegetarian ☐ c. Non-vegetarian ☐
14. If you are non-vegetarian which type of meat do you consume? (*you can choose multiple answers*) a. Fish ☐ b. Beef ☐ c. Chicken ☐ d. Mutton ☐ e. Others ☐	

15. Do you consume tea/coffee? a. Yes ☐ b. No ☐	16. If yes, how often do you drink tea/coffee per day? ……………….

17. Do you consume tea/coffee before taking your main meals? a. Yes ☐ b. No ☐	18. Do you incorporate Vitamin C (Lime, lemon, tomato or tamarind) while cooking green leafy vegetables/meat a. Yes ☐ b. No ☐
19. How often do you consume fruit juices or fruits? a. Never ☐ b. Daily ☐ c. Once a week ☐ d. Once a month ☐	
20. Do you have any of the below chronic diseases? a. Diabetes ☐ b. Heart disease ☐ c. Arthritis ☐ d. Kidney diseases ☐ e. Cancer ☐ f. Liver diseases ☐ g. Other diseases ☐ h. No chronic diseases ☐	

Part II- Reproductive health

If you DO NOT have any children proceed to question number 29

21. How many pregnancies did you have what is par (parity)?………………	22. Number of miscarriage/ abortions? …………………………………..
23. When was your last delivery? ……Date …….. Month ……. Year	24. Was your last pregnancy an/a, a. Institutional delivery ☐ b. Home delivery ☐
25. What is the method of childbirth (delivery a. Normal Vaginal Birth ☐ b. Assisted Vaginal Delivery ☐ c. Caesarean Section ☐	26. Birth spacing (time period between the last two child births): a. 1 year ☐ b. 2-3 years ☐ c. ≥ 4 years ☐
27. Your age during first child birth? ……………	28. What is your status on breast feeding? a. Used to breast feed ☐ b. Currently breast feeding ☐ c. Never nursed a child ☐
29. If you used to breast feed, how long did you breast feed your last child for? ………………..	30. Do you use any contraception methods within the past two months? a. Yes ☐ b. No ☐
31. If yes, specify which of the following contraception methods a. Oral Contraceptive pills ☐ b. IUCD ☐ c. Surgical ☐ d. Others ☐	32. On average how any days does your period last? ……………………
33. Passage of clots during menstruation? a.Yes ☐ b.No ☐	34. Are your menses/periods regular? a.Yes ☐ b. No ☐

Part III - Knowledge on Anaemia

35. Have you heard of anaemia? Yes ☐ No ☐
36. What is the main nutrient that causes anaemia? ……………

Tick **EITHER** True or False for the questions given below

	True	False
37. Worms in the intestine can cause anaemia		
38. Nutrition deficiency the commonest cause of anaemia		
39. Anaemia doesn't affect pregnancy		
40. Diet rich in protein, iron and vitamin C cannot improve anaemia		
41. Anaemia never causes maternal mortality		
42. Green leafy vegetables, drumsticks, beetroot, jiggery are rich sources of iron		
43. Non-vegetarian meals especially red meat and liver good in improving anaemia		
44. Iron supplements are necessary in pregnancy		
45. Anaemia does not cause preterm labour		
46. Anaemia does not cause increased bleeding after delivery		
47. Anaemia can lead to low birth weight babies		
48. Anaemia does not affect the newborn		

Part VI - 24 HOUR dietary recall

Please describe the foods (meals, snacks and drink) that you ate yesterday during the whole day at home or outside the home. Start with the first food/drink taken in the morning.

Write down all foods and drinks mentioned by the respondent. Ask for portion size only for foods rich in carbohydrates and fat. When the respondent has been finished, probe for foods not mentioned.

Meal	Composition
Breakfast	
Morning snack/ tea	
Lunch	
Evening snack/tea	
Dinner	
Other	